

4(a)(i)	A – ciliated epithelial (cell) ; R ciliated epithelium B – goblet (cell) ;	2
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Question	Answer	Marks
4(a)(ii)	1 goblet cells / B, secrete / produce / release, mucus / mucin ; 2 mucus, covers / AW, the (ciliated) epithelium to trap (named) particles ; or <i>idea that</i> mucus prevents (named) pathogens reaching, epithelial cells / alveoli / gas exchange surface ; I lungs A acts as a barrier to (named) pathogens 3 cilia / ciliated epithelial cells, move mucus (and trapped material), upwards / towards mouth / towards throat or pharynx / away from alveoli / away from gas exchange surface / away from lungs / out of the airways ; I 'out of the lungs' / 'to be swallowed' unqualified / out of the respiratory system / out of the gas exchange system	3
4(b)(i)	9+2, pattern / arrangement / structure ; composed of microtubules ; AVP ; e.g. <i>ref. to</i> dynein 'arms' / AW outer 9 are, pairs of microtubules / doublets, and central two are single	2
4(b)(ii)	<i>idea that</i> each cilium / structure X, is surrounded by cell (surface) membrane (so contents are inside the, cell / cytoplasm) ;	1
4(c)	<i>must have either mp1 or mp2 or both to gain max 4, otherwise max 3</i> 1 (centrioles) make / organise, microtubules ; A microtubular organising centre / MTOC 2 to form, the spindle / spindle fibres ; 3 (during each cell cycle) each centriole, replicates / duplicates / forms two centrioles ; 4 during, S phase / G2 phase ; A before mitosis <i>mp4 is linked to mp3</i> 5 centriole pairs / centrioles, move to the (opposite) <u>poles</u> (of cell during prophase of mitosis) ; R if incorrect phase R 'poles of nucleus' 6 centrioles, lengthen / shorten, the spindle fibres / microtubules ; I 'contract'	4